

MIT Academy of Engineering, Alandi

Essentials of Data Science (EDS)

Theory Activity 01

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Problem Statement :-

Formulate 20 problem statements for a given dataset using numpy and Pandas and apply numpy and pandas methods to find the solution for the formulated problem statements.

Dataset :-

Twitter US Airline Sentiment

Below is the required Dataset :-



Tweets.csv

Importing the dataset on python compiler :-

Untitled1.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text

Files

- Analyze your files with code written by Gemini Upload
- sample_data
- Tweets.csv

Disk 70.74 GB available

```
import pandas as pd
import numpy as np

# Load dataset
tweets_df = pd.read_csv('Tweets.csv')
print(tweets_df)
```

	tweet_id	airline_sentiment	airline_sentiment_confidence
0	570306133677760513	neutral	1.0000
1	570301130888122368	positive	0.3486
2	570301083672813571	neutral	0.6837
3	570301031407624196	negative	1.0000
4	570300817074462722	negative	1.0000
...	...	...	...
14635	569587686496825344	positive	0.3487
14636	569587371693355008	negative	1.0000
14637	569587242672398336	neutral	1.0000
14638	569587188687634433	negative	1.0000
14639	569587140490866689	neutral	0.6771

	negativereason	negativereason_confidence	airline
0	NaN	NaN	Virgin America
1	NaN	0.0000	Virgin America
2	NaN	NaN	Virgin America
3	Bad Flight	0.7033	Virgin America
4	Can't Tell	1.0000	Virgin America
...	...	...	...
14635	NaN	0.0000	American
14636	Customer Service Issue	1.0000	American
14637	NaN	NaN	American
14638	Customer Service Issue	0.6659	American

• Problem statements with solutions on numpy :-

1. Find the mean retweet count

```
import numpy as np

tweets_df = pd.read_csv('Tweets.csv')

mean_retweet_count = np.mean(tweets_df['retweet_count'])
print(mean_retweet_count)
```

0.08265027322404371

2. Find the standard deviation of sentiment confidence

```
import numpy as np

tweets_df = pd.read_csv('Tweets.csv')

std_sentiment_conf = np.std(tweets_df['airline_sentiment_confidence'])
print(std_sentiment_conf)
```

0.1628243978712151

3. Find the maximum retweet count

```
import numpy as np

tweets_df = pd.read_csv('Tweets.csv')

max_retweet_count = np.max(tweets_df['retweet_count'])
print(max_retweet_count)
```

44

4. Find the minimum retweet count

```
import numpy as np

tweets_df = pd.read_csv('Tweets.csv')

min_retweet_count = np.min(tweets_df['retweet_count'])
print(min_retweet_count)
```

0

5. Find the median of sentiment confidence

```
import numpy as np

tweets_df = pd.read_csv('Tweets.csv')

median_sentiment_conf = np.median(tweets_df['airline_sentiment_confidence'])
print(median_sentiment_conf)
```

1.0

6. Find the 90th percentile of retweet counts

```
import numpy as np

tweets_df = pd.read_csv('Tweets.csv')

percentile_90_retweets = np.percentile(tweets_df['retweet_count'], 90)
print(percentile_90_retweets)
```

0.0

7. Count the number of tweets having zero retweets

```
import numpy as np

tweets_df = pd.read_csv('Tweets.csv')

zero_retweets_count = np.sum(tweets_df['retweet_count'] == 0)
print(zero_retweets_count)
```

13873

8. Check for outliers in retweet counts using z-score method

```

import numpy as np

tweets_df = pd.read_csv('Tweets.csv')

mean_retweet_count = np.mean(tweets_df['retweet_count'])
z_scores = (tweets_df['retweet_count'] - mean_retweet_count) / np.std(tweets_df['retweet_count'])
outliers_retweets = np.where(np.abs(z_scores) > 3)[0]

print(z_scores)
print(outliers_retweets)

```

```

0      -0.110828
1      -0.110828
2      -0.110828
3      -0.110828
4      -0.110828
...
14635  -0.110828
14636  -0.110828
14637  -0.110828
14638  -0.110828
14639  -0.110828
Name: retweet_count, Length: 14640, dtype: float64
[ 402   445   502  1028  2190  2862  3443  3597  3598  4122  4127  4144
  4464  4902  5270  5550  5551  5607  5995  6485  6486  6858  6969  7013
   7026  7027  7032  7092  7110  7111  7115  7132  7134  7329  7453  8041
   8215  8227  8438  8444  8461  8489  9453  9589  9999 10149 11388 11582
  11591 11596 12428 12675 12676 12677 12678 12680 12681 12684 12783 12848
 14619]

```

9. Find the average retweet count for positive tweets

```

import numpy as np

tweets_df = pd.read_csv('Tweets.csv')

positive_retweet_mean = np.mean(tweets_df[tweets_df['airline_sentiment'] == 'positive']['retweet_count'])
print(positive_retweet_mean)

```

```

0.06940330088870081

```

10. Calculate the correlation between retweet count and sentiment confidence

```

import numpy as np

tweets_df = pd.read_csv('Tweets.csv')

correlation = np.corrcoef(tweets_df['retweet_count'], tweets_df['airline_sentiment_confidence'])[0, 1]
print(correlation)

```

```

0.012580945865928846

```

- Problem Statements with solutions on pandas

1. Find the airline with the most tweets

```
import pandas as pd

tweets_df = pd.read_csv('Tweets.csv')

most_tweets_airline = tweets_df['airline'].value_counts().idxmax()
print(most_tweets_airline)
```

United

2. Find the sentiment distribution (positive, neutral, negative)

```
import pandas as pd

tweets_df = pd.read_csv('Tweets.csv')

sentiment_distribution = tweets_df['airline_sentiment'].value_counts()
print(sentiment_distribution)
```

airline_sentiment

negative	9178
neutral	3099
positive	2363

Name: count, dtype: int64

3. Find the most common negative reason

```
import pandas as pd

tweets_df = pd.read_csv('Tweets.csv')

most_common_negative_reason = tweets_df['negativereason'].value_counts().idxmax()
print(most_common_negative_reason)
```

Customer Service Issue

4. Find the time range of the tweets

```
import pandas as pd

tweets_df = pd.read_csv('Tweets.csv')

time_min = pd.to_datetime(tweets_df['tweet_created']).min()
time_max = pd.to_datetime(tweets_df['tweet_created']).max()
range=time_max-time_min
print(time_max)
print(time_min)
print(range)
```

```
2015-02-24 11:53:37-08:00
2015-02-16 23:36:05-08:00
7 days 12:17:32
```

5. Find the location with the most tweets

```
import pandas as pd

tweets_df = pd.read_csv('Tweets.csv')

most_common_location = tweets_df['tweet_location'].value_counts().idxmax()
print(most_common_location)
```

```
Boston, MA
```

6. Find the top 5 users who tweeted the most

```

import pandas as pd

tweets_df = pd.read_csv('Tweets.csv')

top_5_users = tweets_df['name'].value_counts().head(5)
print(top_5_users)

```

```

name
JetBlueNews      63
kbosspotter      32
_mhertz          29
otisday          28
throthra         27
Name: count, dtype: int64

```

7. Calculate the average sentiment confidence per airline

```

import pandas as pd

tweets_df = pd.read_csv('Tweets.csv')

avg_sentiment_conf_per_airline = tweets_df.groupby('airline')['airline_sentiment_confidence'].mean()
print(avg_sentiment_conf_per_airline)

```

```

airline
American      0.917352
Delta         0.869878
Southwest     0.886516
US Airways    0.921578
United        0.900878
Virgin America 0.876086
Name: airline_sentiment_confidence, dtype: float64

```

8. Find the airline with the most negative tweets

```

import pandas as pd

tweets_df = pd.read_csv('Tweets.csv')

most_negative_airline = tweets_df[tweets_df['airline_sentiment'] == 'negative']['airline'].value_counts().idxmax()
print(most_negative_airline)

```

```

United

```

9. Group by time zone and find the average retweet count


```
import pandas as pd

tweets_df = pd.read_csv('Tweets.csv')

avg_retweets_per_timezone = tweets_df.groupby('user_timezone')['retweet_count'].mean().sort_values(ascending=False)
print(avg_retweets_per_timezone)
```

user_timezone	retweet_count
Pretoria	1.000000
Islamabad	0.500000
Berlin	0.444444
Paris	0.280000
Amsterdam	0.270270
...	...
Tokyo	0.000000
Vienna	0.000000
Warsaw	0.000000
Wellington	0.000000
West Central Africa	0.000000

Name: retweet_count, Length: 85, dtype: float64

10. Find the hour of the day when most tweets were made

```
import pandas as pd

tweets_df = pd.read_csv('Tweets.csv')

tweets_df['tweet_created_dt'] = pd.to_datetime(tweets_df['tweet_created'])
tweets_df['tweet_hour'] = tweets_df['tweet_created_dt'].dt.hour
most_common_hour = tweets_df['tweet_hour'].value_counts().idxmax()

print(most_common_hour)
```

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Thank you!